



UNIVERSITÄT
LEIPZIG

Medizinische Fakultät

Information management in clinical research

MI-Lab Lecture / Exercise

Local Data Hub (LDH)

Leipzig, Date
Mona Perbix

imise.

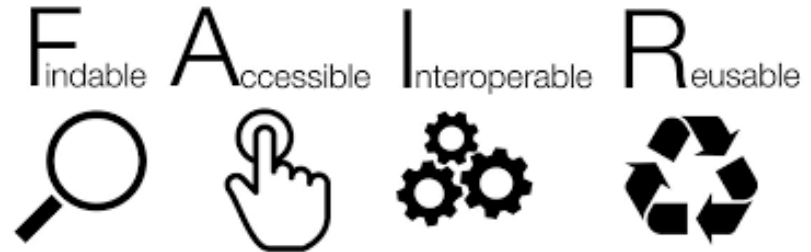
PURPOSE OF THE EXERCISE

- Upload in ein Forschungsdatenmanagementsystem (FDMS)
 - Studiendokumente
 - Datensatz
 - Metadaten
- Ausfüllen von Metadatenschema für eine Studie

INTRODUCTION

WHY UPLOAD TO AN FDMS?

- Data from clinical trials often do not comply with **FAIR principles**
 - Lack of international **standards** for registration and publication
 - Possibilities for **data use** often unclear
 - Restrictions on **reusability**
 - Databases are often not **interoperable**



<https://de.wikipedia.org/wiki/FAIR-Prinzipien>

INTRODUCTION

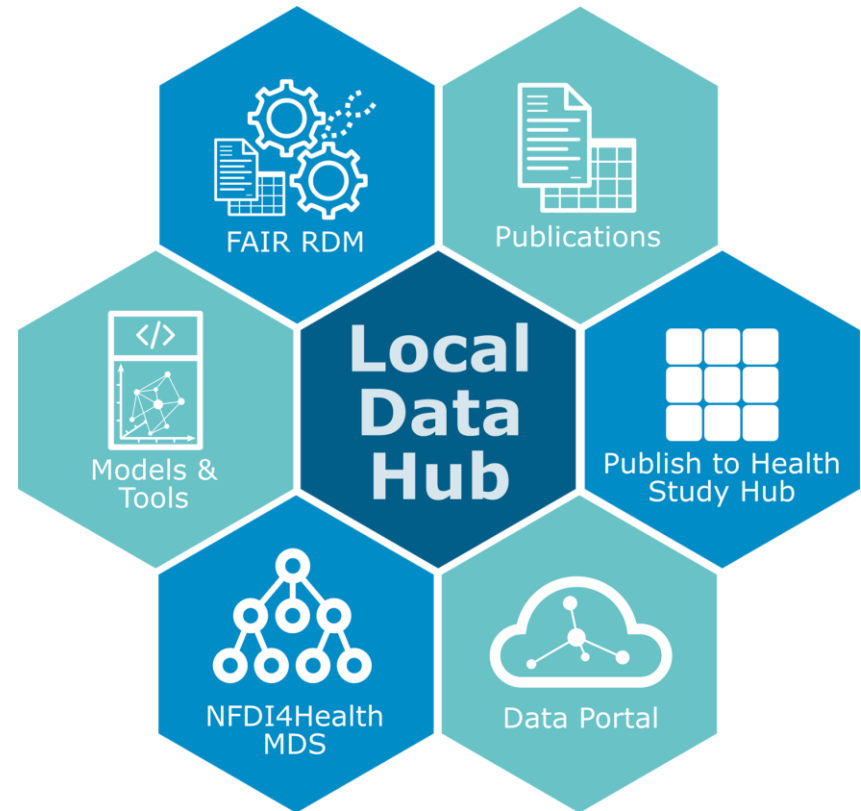
LOCAL DATA HUB (LDH)

- **Open source** software for data management
- Networking of complementary resources from clinical studies
 - Study protocols, data sets, references to publications, etc.
- Project-specific **access controls** and rights assignment
- **Archiving** and versioning
 - Citation & referencing

INTRODUCTION

LDH STRUCTURE

- Direct publication of content in the NFDI4Health **German Central Health Study Hub**
- NFDI4Health metadata schema for standardized publications



<https://www.nfdi4health.de/en/service/local-data-hub.html>

The FAIRDOM SEEK Plattform

Tooling

- FAIRDOM-SEEK is an open source web-based cataloguing and commons platform, for sharing heterogeneous scientific research datasets, models or simulations, processes and research outcomes. It preserves associations between them, along with information about the people and organisations.
- Underpinning FAIRDOM-SEEK is the ISA infrastructure, a standard framework for describing how individual experiments are aggregated into wider studies and investigations. Within FAIRDOM-SEEK, ISA has been extended and is configurable to allow the structure to be used outside of Biology.
- Flexible and detailed sharing permissions are available to manage the catalogued items from early collaborations within projects, through to the publishing of final research results

<https://www.nfdi4health.de/en/service/local-data-hub.html>

INTRODUCTION

INVESTIGATION, STUDY, ASSAY (ISA) FRAMEWORK

1. Investigation

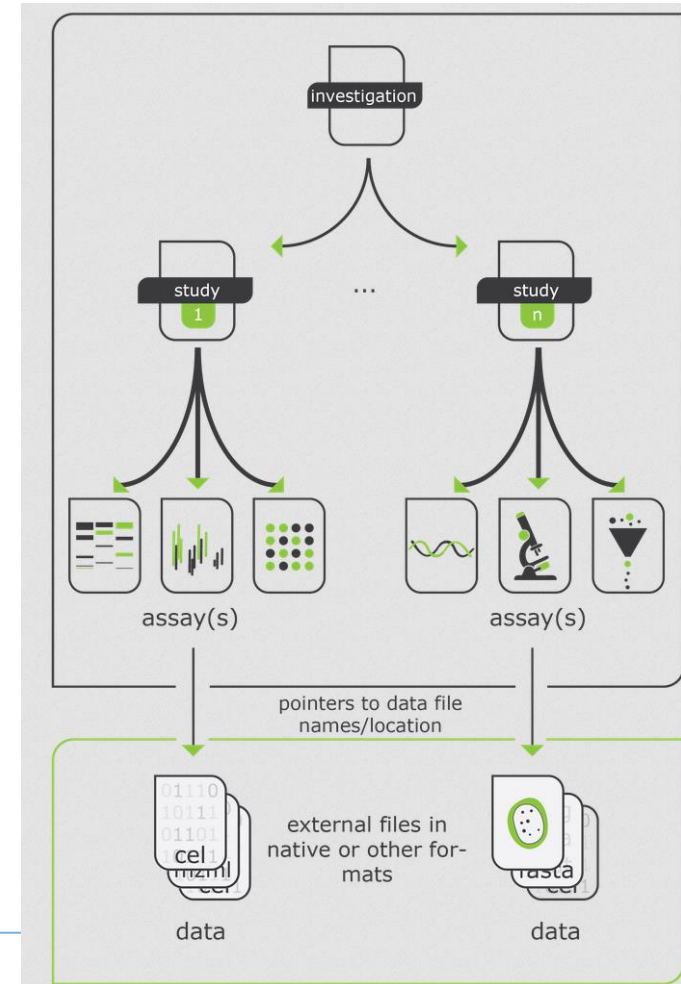
- High level concept
- Links related studies

2. Study

- Central unit
- Contains information on the subject
- Has associated assays

3. Assay

- Tests performed which produce data



LDH CASE STUDY

Name of the study

Related investigation and project

Files for download

The screenshot shows the 'imise' web interface. At the top, there is a search bar and navigation links. The main content area displays the 'HNSCC' study. A red arrow points to the 'HNSCC' title, which is labeled 'Name of the study'. Below the title, there are tabs for 'Overview' and 'Related Items'. A red bracket highlights the 'Investigation' and 'Projects' sections, labeled 'Related investigation and project'. Another red bracket highlights the 'Files for download' section, which lists various data files like 'HNSCC OMICS Data', 'Gene expression patterns and TP53 mutations...', 'HNSCC Clinical Dataset', 'GSE 65858 XLS', 'GSE 65858 ODM', and 'The role of HPV RNA transcription...'. The URL at the bottom is <https://datrav.uni-leipzig.de/ldh-milab/studies/1>.

Why Extended Metadata?

- Resources only have a few fixed attributes:

SEEK ID: <https://datrav.uni-leipzig.de/ldh-milab/studies/2>

Investigation: [Medical Information Mart for Intensive Care \(MIMIC\)-IV](#)

Projects: [MI-Lab](#)

Study position:

- **Goal:** Add your own metadata attributes
- Summarized as **a metadata schema** for easy import

NFDI4HEALTH METADATA SCHEMA

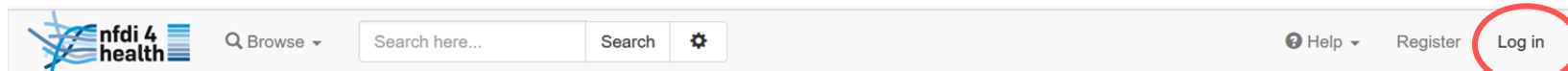
- For standardized publication on the German Central Health Study Hub
 - Study resources
 - Metadata
- FAIR publication of health studies

DOCUMENTS TO BE UPLOADED

Document	Description
Research Data Management Plan (FDMP)	How research data is created, processed, stored, and shared
Study protocol	Contains the objective, design, and methodology of the study
Data quality rules	Standards and criteria for data quality
Data Dictionary	Structure and variables of the data set. Facilitates interpretation of the data

See GitHub: [Material/Study_documents](#)

REGISTRATION LDH



University of Leipzig - Research Data Management Hub: The better way to manage your data

Free and open platform for easier research data management



Organise & Structure

Organise documents and data to easily manage your research project



Manage access

There is a lot of flexibility and control over who can see, download or edit your items



Standardise research data

Use available standards to describe your dataset and make it ready for publication



Publish & Share data

Publish your data and link them to your scientific article

Learn more

<https://ldh.imise.uni-leipzig.de/>

Register

Exercise: GitHub

GitHub: [MI-Lab-E04-FDM](#)

Exercise sheet under /Exercise



Exercise: Task 1

Institution and project

Search for the institution "Institute for Medical Informatics, Statistics and Epidemiology (IMISE)".

Question: Which projects are listed under this institution?

Exercise: Task 2.1

Creating an investigation

Add a new investigation called "Medical Information Mart for Intensive Care (MIMIC)-IV" to the "MI-Lab" project.

Extended metadata schema

Select the extended metadata schema **"MIAPPE metadata v1.1."**

Add the required metadata for the investigation.

Exercise: Task 2.2

Access rights

Configure the access rights for this investigation so that only people from your project can access it.

Questions

1. Why are there access rights for clinical studies?
2. Why should you not make all data from a clinical trial openly available?

Exercise: Task 3

Creating a study

Add a new study called "MIMIC IV Study" under the investigation you have already created.

3.1 Extended metadata schema

Select the extended metadata schema "**MIMCT Metadata V0.7 for object type Study**" and add the required metadata.

3.2 Access rights

Define the access rights for the study so that only people from your project can access it.

Exercise: Task 4.1

Creating an experimental assay

Add a new experimental assay called "MIMIC IV Clinical Dataset" to your study. Then upload the following files that are available for this study:

4.1 Publications

Click Create and add several publications. The PubMed IDs are:

36596836 and **36688534**

Link the publications to the study you have already created.

Exercise: Task 4.2

Documents

Add several documents to the experimental assay you have already created. These can be found in **GitHub** under "Material/Study_documents":

- MIMIC IV FDMP
- MIMIC IV study protocol
- MIMIC IV Data Quality Rules
- MIMIC IV Data Dictionary

Edit the access rights for the files.

Questions:

1. Why should these documents be uploaded in addition to the data set?
2. What are these documents used for in a clinical study?

Exercise: Task 4.3

Data

Add the MIMIC IV data set to the experimental assay you have already created. This can be found in **GitHub** under "Material/Dataset":

MIMIC_IV_Dataset.csv



UNIVERSITÄT
LEIPZIG

Medizinische Fakultät

THANK YOU!

Mona Perbix

Institut für Medizinische Informatik, Statistik und Epidemiologie (IMISE)

www.imise.uni-leipzig.de